

University of Thessaly
Department of Electrical and Computer Engineering

Information Retrieval Assignment

ECE328 — Spring 2026

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1. Part 1 — Link Analysis (4 points)

1.1. Part 1A — Personalised PageRank on the 14-node graph

Setup

We computed Personalised PageRank on the 14-node undirected graph shown in Figure 1, using damping factor $a = 0.65$. The teleportation vector $\mathbf{v}$ is non-uniform: node 14 receives probability mass 0.5, and the remaining 0.5 is distributed uniformly across the other 13 nodes ($\approx 3.85\%$ each). All edges are bidirectional; the graph has 22 edges forming three loosely connected clusters: a left cluster $\{1, 2, 3, 4, 5\}$, a bottom cluster $\{6, 7, 8, 9\}$, and a right cluster $\{10, 11, 12, 13, 14\}$, joined by single-edge bridges.

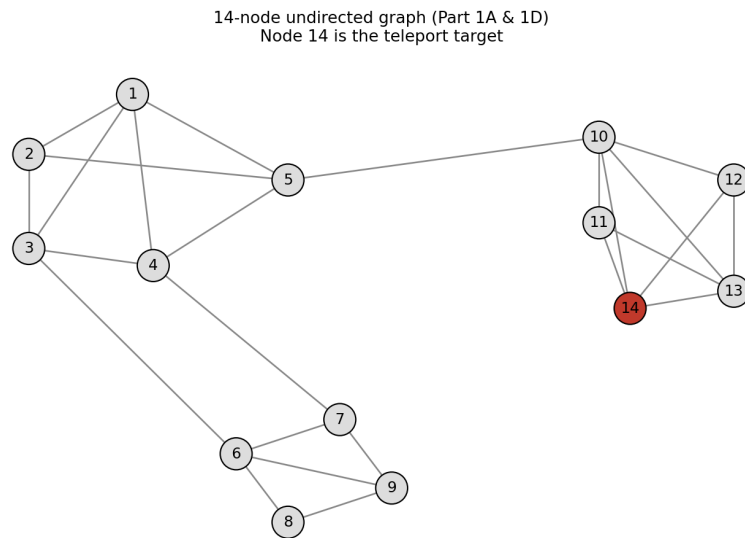


Figure 1: The 14-node undirected graph used in Parts 1A and 1D. Node 14 (highlighted in red) is the teleport target.

Results

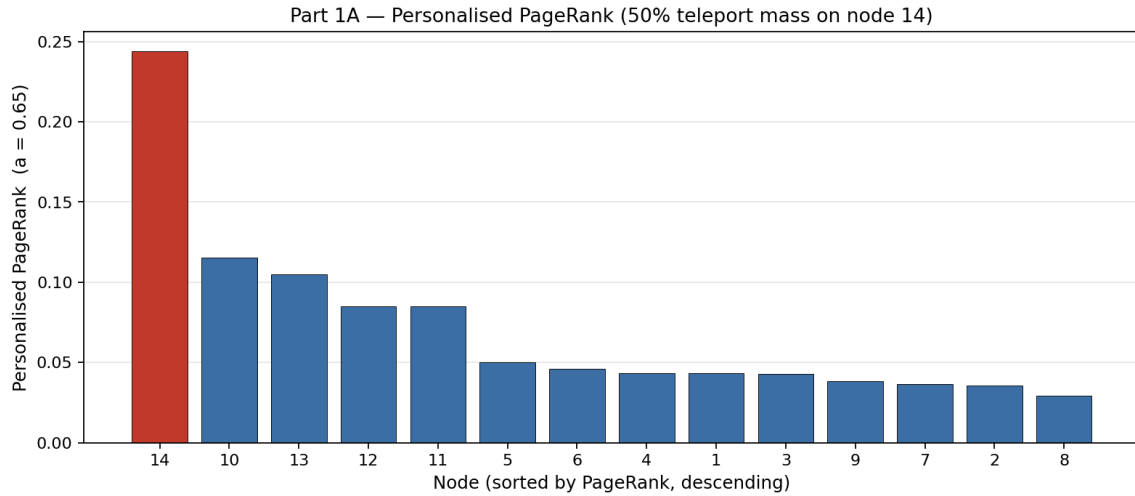


Figure 2: Personalised PageRank scores for all 14 nodes (sorted descending). Node 14 is highlighted in red.

The top-5 ranked nodes are reported in Table 1.

Rank	Node	PageRank Score	Degree
1	14	0.243929	4
2	10	0.115221	5
3	13	0.104971	4
4	12	0.085136	3
5	11	0.085136	3
6	5	0.050279	4
7	6	0.046101	4
8	4	0.043505	4
9	1	0.043376	4
10	3	0.042792	4
11	9	0.038333	3
12	7	0.036328	3
13	2	0.035634	3
14	8	0.029258	2

Table 1: PageRank values with personalization on node 14 ($\alpha = 0.65$)

Discussion

Node 14 ranks first by a large margin (score 0.2439, more than double the second-ranked node), solely due to the 50% teleportation mass assigned to it. Node 10 occupies the second position (score 0.1152), acting as the crucial gateway between the right cluster and the rest of the graph. The other immediate neighbours in the right cluster—nodes 13, 12, and 11—occupy the next three positions, since random walks that restart on node 14 immediately propagate probability mass to them.

Critically, the personalisation vector overrides the topological signal: in standard uniform-teleport PageRank, well-connected hub nodes such as node 4 (degree 4) and node 1 (degree 4) would rank at the top; here they fall below the entire right cluster because the repeated injections of probability mass near node 14 flood that region.

Nodes in the bottom cluster ($\{6, 7, 8, 9\}$) rank last, consistent with their greater graph distance from node 14. Node 8 in particular has degree 2 (connected only to nodes 6 and 9) and a long shortest-path distance to node 14, producing the lowest score overall.

1.2. Part 1B — Damping factor sweep on the 8-node directed graph

Setup

We computed PageRank on the 8-node directed graph (Figure 3) for five damping factors $a \in \{0.55, 0.65, 0.75, 0.85, 0.95\}$. Each configuration uses a uniform teleportation vector. We then measured the stability of the induced rankings by computing pairwise Kendall τ between all $\binom{5}{2} = 10$ pairs of rankings.

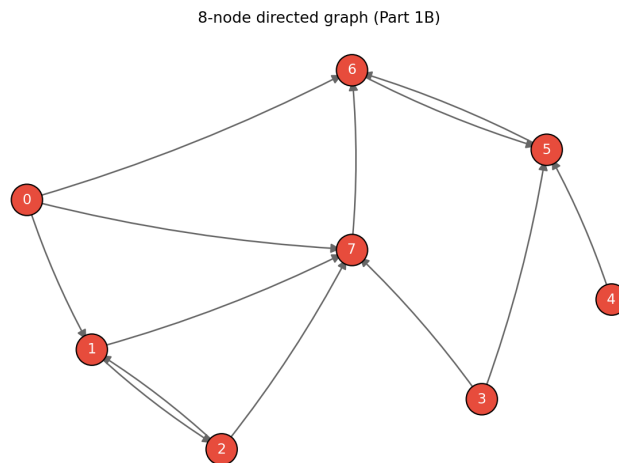
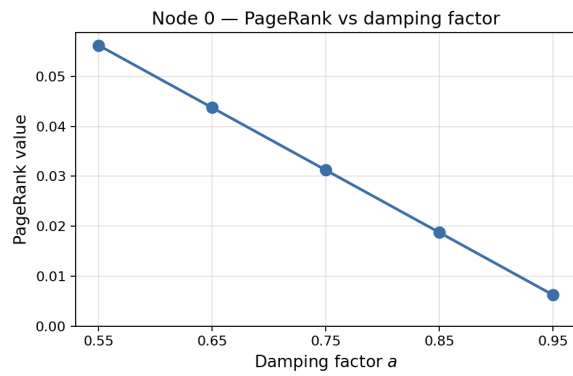


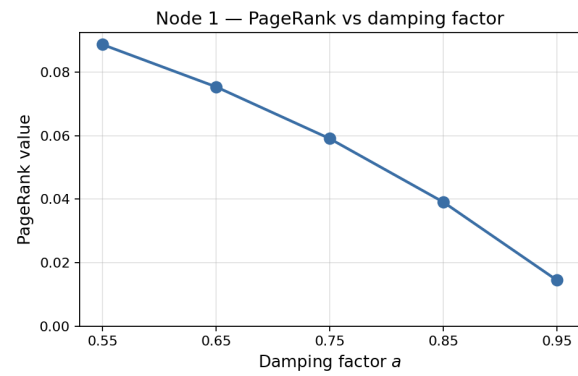
Figure 3: The 8-node directed graph used in Part 1B.

Results

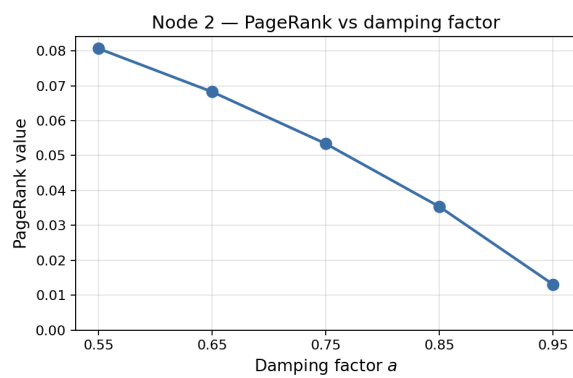
Figures 4a–5d show the PageRank value of each node as a function of the damping factor a , one figure per node.



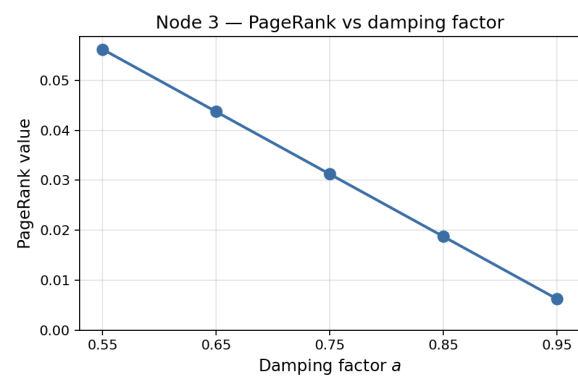
(a) Node 0



(b) Node 1



(c) Node 2



(d) Node 3

Figure 4: Part 1B — PageRank vs damping factor, nodes 0–3.

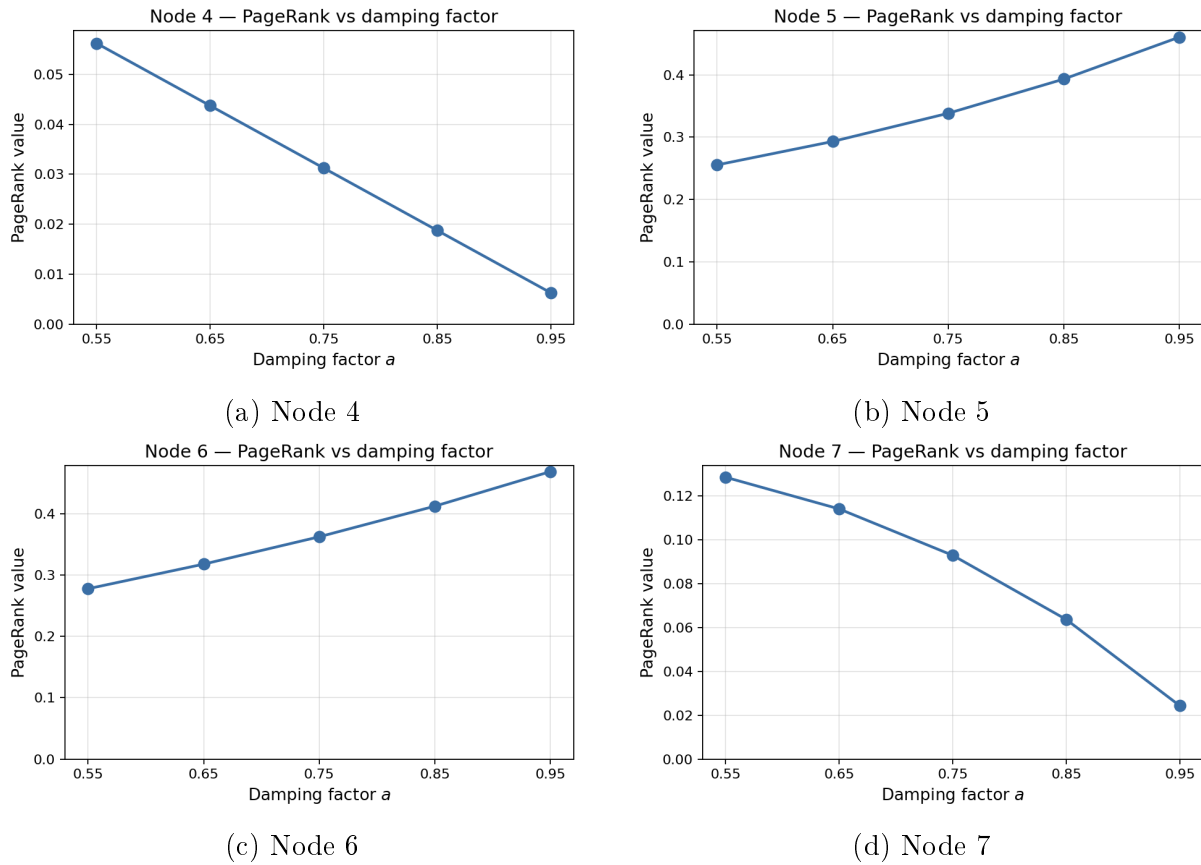
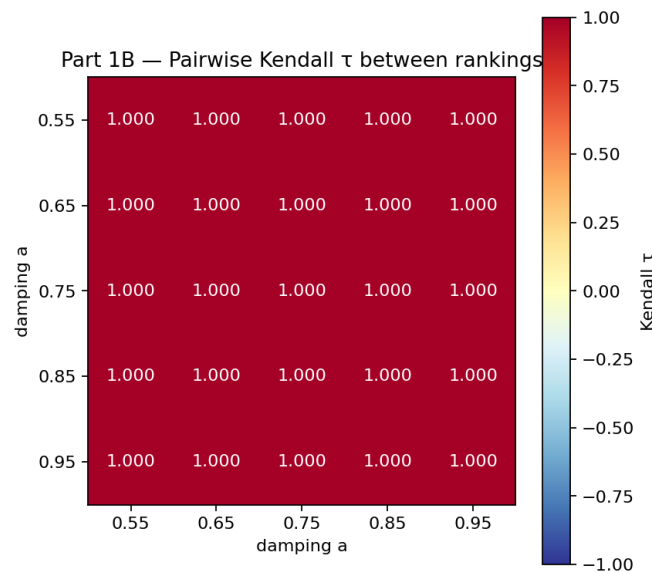


Figure 5: Part 1B — PageRank vs damping factor, nodes 4–7.

Figure 6: Pairwise Kendall τ between the five rankings produced at different damping values. All entries are 1.00.

Discussion

From Figures 4–5, the absolute PageRank values change substantially as a increases: nodes 5 and 6—which together form a sink strongly absorbing random-walk probability

(they exchange edges between each other and receive edges from most other nodes)—see their scores rise sharply toward $a = 0.95$, while all other nodes, whose mass leaks toward 5 and 6, approach zero. This behaviour is characteristic of nearly-dangling sink clusters: at high damping the walk spends nearly all its time in the strongly connected component $\{5, 6\}$.

However, the *ranking order* is perfectly stable: every pairwise Kendall $\tau = 1.00$ (Figure 6). This means no two nodes swap positions for any value of a in $[0.55, 0.95]$. The relative importance hierarchy is determined entirely by graph topology and is invariant to the damping parameter over this range.

1.3. Part 1C — HITS on the third graph

Setup

We applied the HITS algorithm [4] to the 8-node directed graph shown in Figure 7, whose nodes are labelled $\{2, 3, 5, 7, 8, 9, 10, 11\}$. Authority scores are the principal eigenvector of $A^T A$; hub scores are the principal eigenvector of AA^T . Both matrices are real symmetric positive semi-definite, so we used `numpy.linalg.eigh` and took the eigenvector corresponding to the largest eigenvalue. Results were cross-checked against `networkx.hits`.

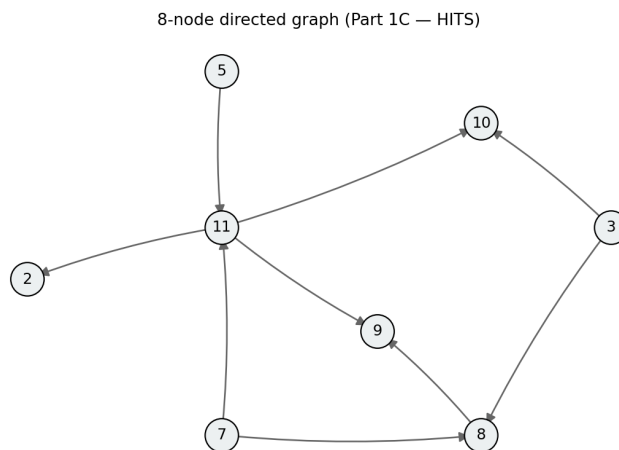


Figure 7: The directed graph used in Part 1C (HITS).

Results

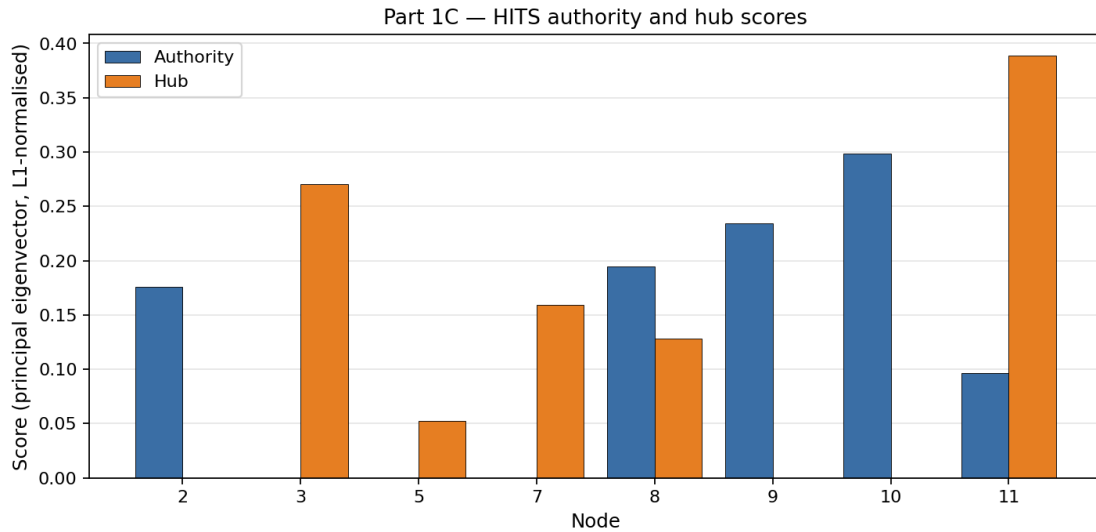


Figure 8: HITS authority and hub scores for each node (L1-normalised principal eigenvectors of $A^T A$ and AA^T respectively).

Table 2: HITS scores (L1-normalised principal eigenvectors).

Node	Authority	Hub
2	0.1761	0.0000
3	0.0000	0.2706
5	0.0000	0.0527
7	0.0000	0.1595
8	0.1948	0.1284
9	0.2343	0.0000
10	0.2987	0.0000
11	0.0961	0.3887

Discussion

Authority and hub scores capture complementary structural roles. **Node 10** is the top authority (score 0.299): it receives edges from node 11 and node 3, both of which are strong hubs, and these hubs point to it specifically. **Node 11** is the top hub (score 0.389): it has the highest out-degree in the graph (edges to nodes 2, 9, and 10) and is pointed to by both node 5 and node 7.

Nodes with zero hub score (2, 9, 10) have no outgoing edges and are pure sinks; nodes with zero authority score (3, 5, 7) have no incoming edges from other hubs and are pure sources. Node 8 is the only node that plays both roles meaningfully (authority 0.195, hub 0.128): it both receives an edge from node 3 (a hub) and emits an edge to node 9 (boosting its hub score).

1.4. Part 1D — Eigenvalues of the Google matrix

Setup

We computed the eigenvalues of the Google matrix

$$G = aM + \frac{1-a}{n}\mathbf{1}\mathbf{1}^\top$$

for the 14-node graph at $a = 0.85$, where M is the row-stochastic transition matrix. We then added the five cross-cluster edges (5–11), (4–11), (7–13), (8–12), (9–13) and recomputed the eigenvalues of the modified graph.

Results

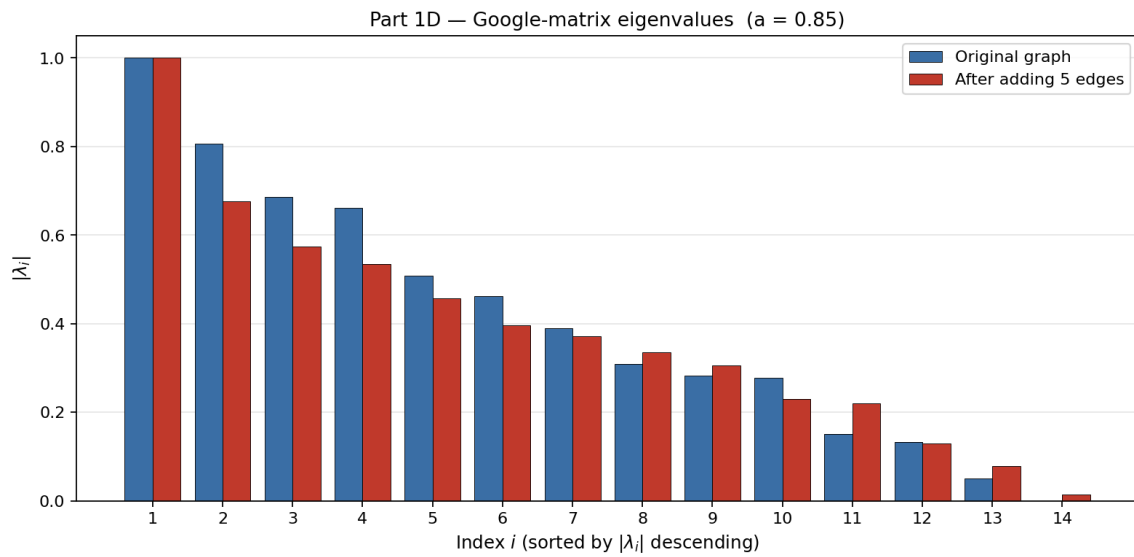


Figure 9: Magnitudes of the Google-matrix eigenvalues before and after adding five cross-cluster edges ($a = 0.85$).

Table 3: Key spectral quantities before and after edge addition.

Quantity	Original	Modified
$ \lambda_1 $	1.0000	1.0000
$ \lambda_2 $	0.8071	0.6769
Spectral gap ($1 - \lambda_2 $)	0.1929	0.3231
Number of edges	22	27

Discussion

The dominant eigenvalue $\lambda_1 = 1$ is invariant in both cases: any row-stochastic matrix has spectral radius exactly 1.

The key quantity is $|\lambda_2|$, which controls the convergence rate of the power iteration used to compute PageRank: the error after t iterations decays as $\mathcal{O}(|\lambda_2|^t)$, so a smaller $|\lambda_2|$ means faster convergence.

Original graph. The three weakly-coupled clusters connected by single-bridge edges produce an $|\lambda_2| = 0.807$, close to the theoretical ceiling of $a = 0.85$. The Markov chain mixes slowly because a random walk can be “trapped” in one cluster for many steps.

Modified graph. Adding five cross-cluster edges reduces $|\lambda_2|$ to 0.677, and the spectral gap nearly doubles from 0.193 to 0.323. The new edges directly connect the previously isolated clusters, creating shorter paths between all pairs of nodes and making the stationary distribution easier to reach from any starting point. The remaining eigenvalues $|\lambda_3|, \dots, |\lambda_{14}|$ also compress uniformly downward (visible in Figure 9), reflecting the more uniform mixing geometry of the modified graph.

2. Part 2 — Zipf’s Law on *Pride and Prejudice* (1 point)

2.1. Setup

We tested Zipf’s law [6] on Jane Austen’s *Pride and Prejudice* (Project Gutenberg eBook #1342, Plain Text UTF-8 [1]). The text was stripped of Project Gutenberg boilerplate and tokenised with the regex `[a-z]+(?:'[a-z]+)?`, which preserves contractions as single tokens (e.g. *don’t*) and lower-cases all words. **No stemming** was applied.

Zipf’s law states that the collection frequency of the i -th most frequent term satisfies

$$\text{cf}_i \approx \frac{c}{i},$$

i.e. in log-log space the frequency–rank curve is a straight line with slope -1 .

The constant c was estimated by three methods:

1. $c_a = \text{cf}_1$ (the most-frequent term’s count).
2. $c_b = \text{median}(\text{cf}_i \cdot i)$ across all ranks.
3. c_c : least-squares fit of $\log \text{cf}_i = \log c - b \log i$ over the top 1000 ranks.

Estimator (a) is the primary result, derived directly from the boundary condition $i = 1 \Rightarrow c = \text{cf}_1$; estimators (b) and (c) are provided as cross-checks. The log-log fit also yields the empirical exponent b (Zipf predicts $b = 1$).

2.2. Results

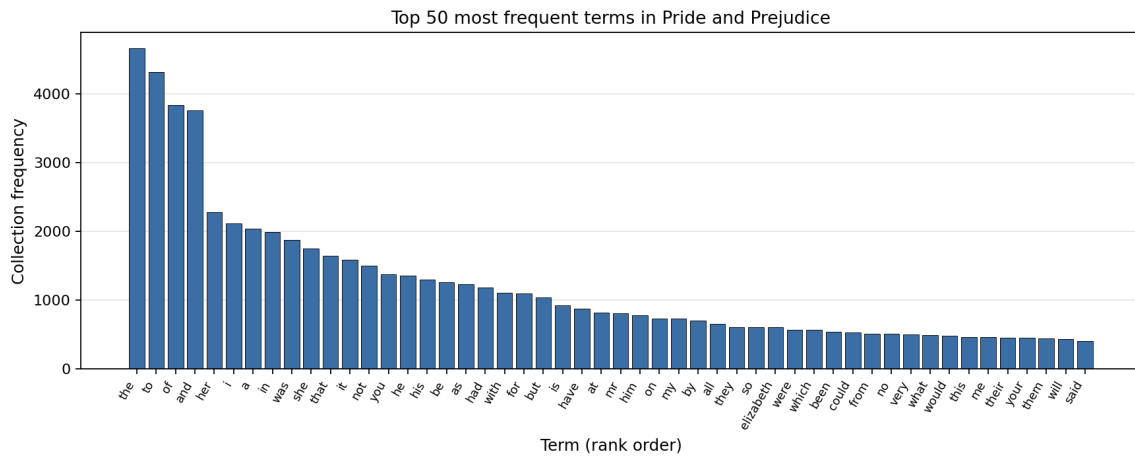


Figure 10: Collection frequencies of the 50 most frequent terms in *Pride and Prejudice*.

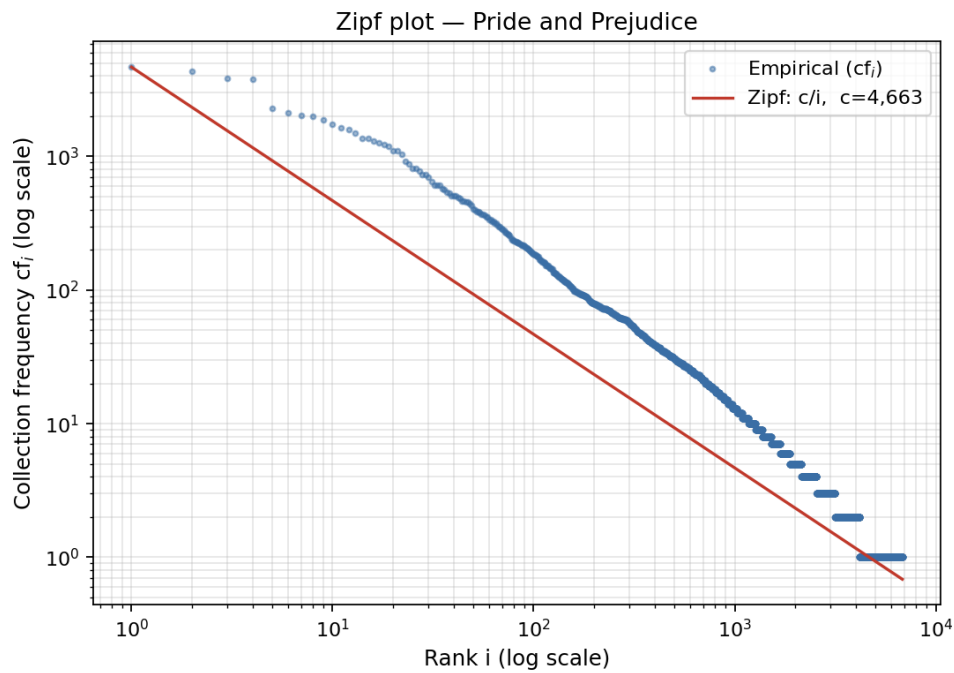


Figure 11: Log-log Zipf plot. Blue dots: empirical frequencies. Red line: Zipf model c/i , $c = 4663$.

Table 4: Estimates of the Zipf constant c for *Pride and Prejudice*.

Estimator	c
(a) cf_1 ("the")	4 663
(b) $\text{mean}(cf_i \cdot i)$	8 734
(b) $\text{median}(cf_i \cdot i)$	7 788
(c) log-log fit (top 1 000 ranks)	24 874
Empirical exponent b	1.080

2.3. Discussion

Zipf’s law is clearly confirmed for this corpus. The log-log plot (Figure 11) exhibits a near-straight decay over roughly two decades of rank, and the fitted exponent $b = 1.080$ is within 8% of Zipf’s theoretical prediction of 1.0.

The primary constant $c = 4663$ follows directly from the definition: setting $i = 1$ gives $c = cf_1$, the observed frequency of “the”. The higher values produced by estimators (b) and (c) are expected: the log-log slope of -1.08 is slightly steeper than -1 , so the fitted line extrapolates to a larger intercept than the actual peak frequency. This is a well-documented head effect (cf. the Mandelbrot correction), and does not indicate inconsistency among the estimators.

The top-50 bar chart (Figure 10) shows the characteristic sharp falloff after the first handful of function words (*the, to, of, and, her*). The appearance of “elizabeth” at rank 29 and “mr” at rank 23 is domain-specific: these are character references, not generic function words, and their relatively high frequency reflects the novel’s focus on the Bennet family and Mr. Darcy.

Deviations from the model are concentrated in two regions: (i) the *head* (ranks 1–15), where function words are slightly below the fit due to the Mandelbrot effect, and (ii) the *tail*, where hapax legomena (words appearing exactly once) form a flat staircase that falls well below the extrapolated line.

3. Part 3 — Bayesian Personalised Ranking (5 points)

3.1. Setup

We implemented Bayesian Personalised Ranking (BPR) [5] for top- k recommendation on two datasets:

- **Last.FM** (HetRec 2011 [2]): 1892 users, 17 632 artists, 92 834 user–artist interactions. Any listening event is treated as a positive signal.
- **MovieLens 1M** [3]: 6 040 users, 3 706 movies, 1 000 209 ratings. Any rating (regardless of value) is treated as a positive interaction.

Split: per user, 80% of interactions are kept for training and 20% are held out for testing. Users with fewer than 2 interactions remain entirely in the training set.

Evaluation protocol: each experiment is repeated **10 times** with different random seeds for the train/test split. Results report **mean** $\pm$ **std** over these 10 runs.

Implementation: BPR matrix factorisation from the `implicit` library (v0.7.2, Cython backend), with `iterations=100`, `learning_rate=0.01`, `regularization=0.01`.

Metrics:

$$\text{Precision@}k = \frac{|\text{recommended} \cap \text{relevant}|}{k}, \quad \text{Recall@}k = \frac{|\text{recommended} \cap \text{relevant}|}{|\text{relevant}|}.$$

3.2. Part 3A — Effect of latent factors (top-10 recommendation)

We fixed $k = 10$ and varied the number of latent factors $f \in \{10, 20, \dots, 100\}$.

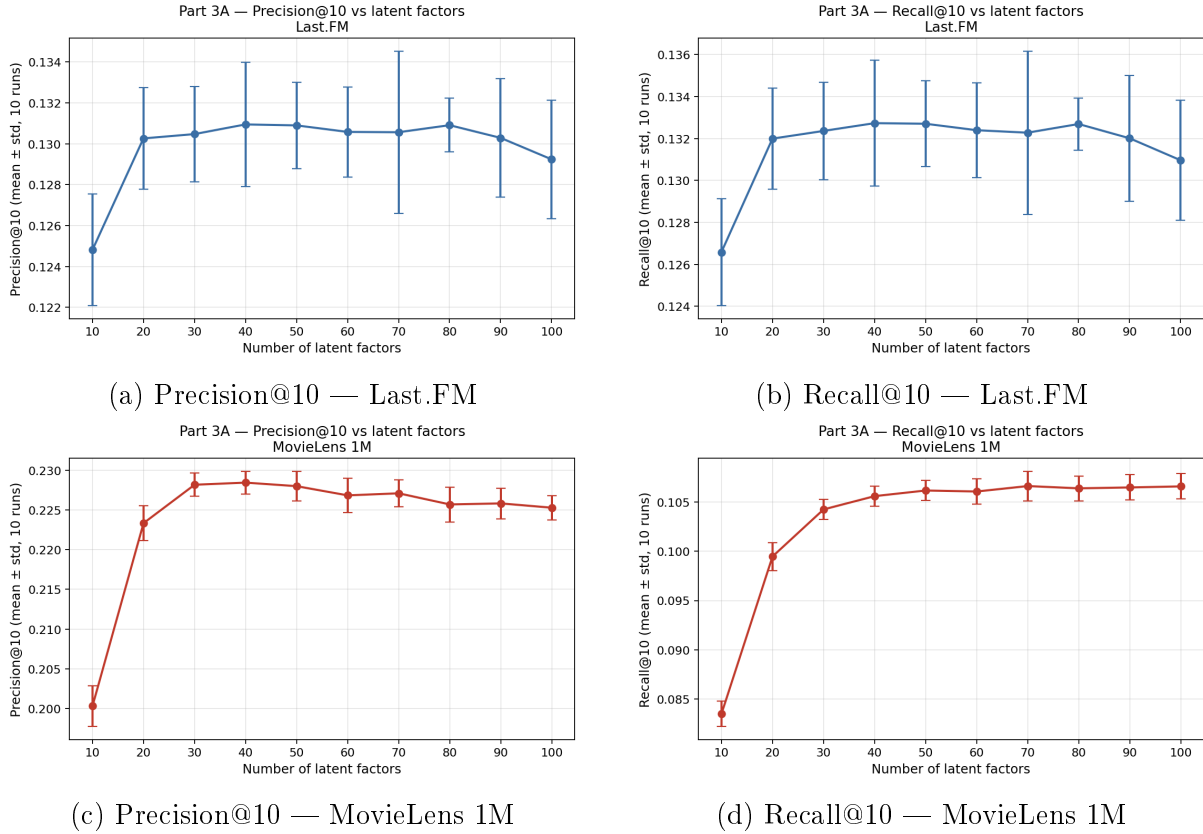


Figure 12: Part 3A: Precision@10 and Recall@10 as a function of the number of latent factors ($k = 10, 10$ runs each, error bars = ± 1 std).

Discussion

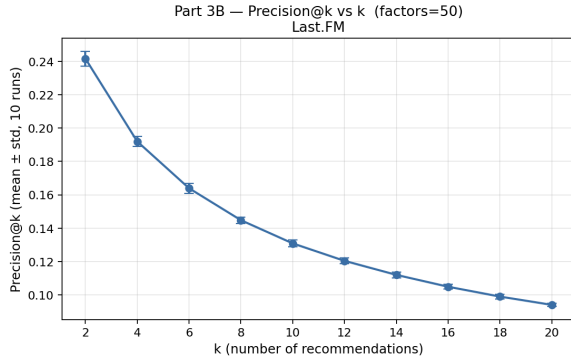
Both datasets show a fast initial gain from $f = 10$ to $f = 30$, after which performance plateaus and then very slightly declines toward $f = 100$. The saturation indicates that the preference space of these datasets is well-captured by ≈ 30 latent dimensions; adding more factors does not improve generalisation and may introduce mild overfitting to training interactions.

MovieLens 1M consistently outperforms Last.FM on Precision@10 (peak ≈ 0.228 vs ≈ 0.131) because it is a denser dataset with more interactions per user, giving the model more signal to learn from. On Recall@10 the gap is inverted: Last.FM achieves higher recall (≈ 0.133 vs ≈ 0.106) because its item catalogue is larger relative to the number of users, so the 20% held-out set contains more items and each correctly recommended item yields a larger recall gain.

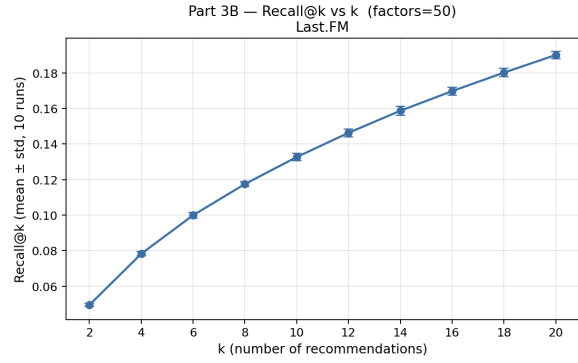
The standard deviations across 10 runs are consistently small (below 0.004), confirming that the results are stable and not sensitive to the particular train/test split.

3.3. Part 3B — Effect of k (fixed factors = 50)

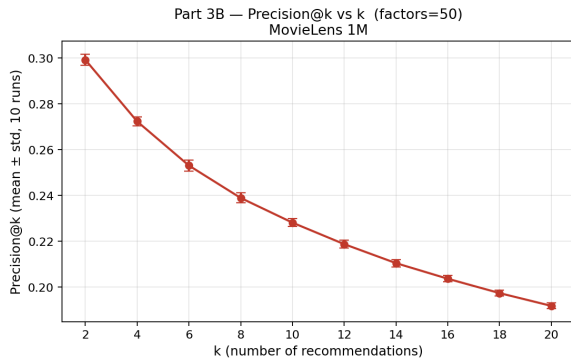
We fixed the number of latent factors to 50 and varied $k \in \{2, 4, 6, \dots, 20\}$.



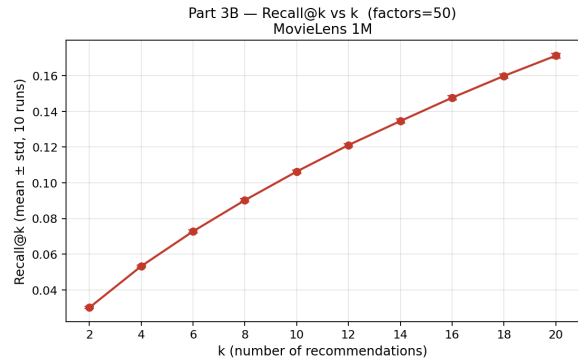
(a) Precision@k — Last.FM



(b) Recall@k — Last.FM



(c) Precision@k — MovieLens 1M



(d) Recall@k — MovieLens 1M

Figure 13: Part 3B: Precision@k and Recall@k as a function of k (factors=50, 10 runs each, error bars = ± 1 std).

Discussion

The results exhibit the classic precision–recall trade-off:

Precision@k decreases monotonically as k grows. At $k = 2$, the model recommends only its two most confident items, so the hit rate is high (Last.FM: 0.242; MovieLens 1M: 0.299). As k increases, the list is padded with less-confident recommendations, diluting precision.

Recall@k increases monotonically with k . A longer list captures more of the held-out items, improving recall. At $k = 20$, Last.FM recall reaches 0.190 and MovieLens 1M reaches 0.171.

MovieLens 1M maintains higher precision than Last.FM across all k values, consistent with the denser interaction matrix giving the model better estimates of user preferences. Last.FM achieves higher recall at larger k , consistent with the explanation given in Part 3A (larger item space, fewer interactions per user, more diverse held-out sets).

The error bars remain very tight throughout, confirming the robustness of the 10-repetition protocol.

4. Conclusions

This report covered three topics in information retrieval.

In **Part 1** we studied link-analysis algorithms on small graphs. Personalised PageRank concentrates importance near the teleport target, overriding the topological hierarchy. The PageRank ranking on the directed graph is perfectly stable across all tested damping factors ($\tau = 1$ for all pairs), while the absolute scores shift substantially. HITS cleanly separates hub and authority roles, identifying nodes that either point to many good sources or are pointed to by many good hubs. Finally, adding cross-cluster edges to a weakly-connected graph nearly doubles the spectral gap of the Google matrix, accelerating the convergence of power iteration.

In **Part 2** we confirmed Zipf’s law on *Pride and Prejudice*. The empirical exponent $b = 1.080$ is within 8% of the theoretical prediction of 1.0, and the log-log plot is approximately straight over two decades of rank. Deviations at the head and tail are consistent with the known Mandelbrot correction and the discrete nature of hapax legomena.

In **Part 3** we applied BPR to implicit-feedback recommendation on two real datasets. Performance saturates at roughly 30 latent factors on both datasets, suggesting that the user-preference space is low-dimensional. The precision–recall trade-off as k varies is clean and monotone, confirming that BPR produces well-calibrated rankings.

Software. All experiments were implemented in Python 3.12 using `networkx` (graph algorithms), `numpy/scipy` (linear algebra), `matplotlib` (figures), `pandas` (tabulation), and `implicit` (BPR). All code is available at <https://github.com/mavroulis/ir-assignment-2026>.

References

- [1] Jane Austen. *Pride and prejudice*. Project Gutenberg eBook #1342, 1813. Plain Text UTF-8 edition.
- [2] Iván Cantador, Peter Brusilovsky, and Tsvi Kuflik. Second workshop on information heterogeneity and fusion in recommender systems (HetRec 2011). In *Proceedings of the 5th ACM Conference on Recommender Systems (RecSys)*, pages 387–388, New York, NY, 2011. ACM.
- [3] F. Maxwell Harper and Joseph A. Konstan. The MovieLens datasets: History and context. *ACM Transactions on Interactive Intelligent Systems*, 5(4):19:1–19:19, 2015.
- [4] Jon M. Kleinberg. Authoritative sources in a hyperlinked environment. *Journal of the ACM*, 46(5):604–632, 1999.
- [5] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. BPR: Bayesian personalized ranking from implicit feedback. In *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 452–461, 2009.
- [6] George Kingsley Zipf. *Human Behavior and the Principle of Least Effort*. Addison-Wesley, Cambridge, MA, 1949.